

TEJASWI REDDY VALLAPU

Machine Learning Engineer | Generative AI Systems

Denton, Texas, USA

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EDUCATION

[University of North Texas](#) | **Master of Computer and Information Sciences**

Denton, Texas

Courses: Machine Learning, Deep Learning, NLP, Distributed Systems, Cloud Computing

Aug 2023 – May 2025

[Jawaharlal Nehru Technological University](#) | **Bachelor of Information Technology**

Hyderabad, India

Courses: Data Structures, DBMS, Operating Systems, Computer Networks, Software Engineering

Jun 2018 – Jul 2022

EXPERIENCE

Machine Learning Engineer @ JP Morgan Chase & Co., US

Jul 2025 – Present

- Deployed **AI-driven financial analytics** on **Azure ML** and **Databricks** using transformers, RAG, and Azure OpenAI to automate portfolio and compliance insights, cutting review time by **60%**, **saving** 15+ hours weekly per analyst
- Built **fraud-detection** and **credit-risk models** using **XGBoost**, **PyTorch Lightning**, and **Azure AutoML**, improving precision by **25%** while embedding **explainable AI** components - SHAP and LIME for enhanced audit transparency.
- Implemented **containerized ML microservices** using **Docker** and **Kubernetes** on Azure, integrated with **Event Hubs** and **REST APIs** for real-time model inference, achieving **89% uptime** in production.
- Engineered **ETL pipelines** via **Azure Data Factory** and **Delta Lake** to process **50M+ financial records daily**, ensuring high data reliability, lineage tracking, and end-to-end automation across analytics workflows.
- Designed a **robust RAG-based Q&A chatbot** leveraging **LangChain** and **Gradio** by integrating **LLMs** with a **vector database**, enabling natural-language querying across **200+ complex datasets**.
- Achieved a **60% boost** in query-resolution speed and a **50% gain** in retrieval precision, saving the data team **10+ hours weekly** using **Chain-of-Thought** and **Few-Shot Prompting** techniques.

Machine Learning Engineer Intern @ MetLife, US

Dec 2024 – Apr 2025

- Designed a constrained risk-scoring framework that decouples severity and frequency models for insurance claims prediction, enabling **formal actuarial compliance guarantees** across underwriting pipelines.
- Integrated **Bayesian** uncertainty quantification with **Student's t-based confidence intervals** to enforce probabilistic reliability thresholds and provide **formal performance guarantees** during model validation and evaluation.
- Fine-tuned **Gradient Boosting** and deep **neural network** ensembles for claims **triage** and **fraud scoring**, improving **inference reliability by 15%**; deployed reproducible **ML pipelines** with auditable **validation tests** for insurance compliance.
- Automated preprocessing pipelines including covering feature engineering, data validation, and train-test preparation across **2 datasets**, **cutting preprocessing effort by 35%**.
- Deployed lightweight **FastAPI + Docker** inference pipelines for internal ML model serving, eliminating manual model handoffs and **reducing validation time by 40%**.

ML Engineer / Associate Software Engineer @ Accenture, India

Jan 2023 – Jun 2023

- Enhanced sales-to-forecast alignment in the **Snowflake** data warehouse using **multi-objective loss optimization** and feature normalization to improve downstream tasks such as pricing prediction, demand forecasting, and revenue analytics.
- Explored 10+ feature engineering variants using contrastive learning schemes, achieving **6.2% improvement in forecast accuracy** and **9.6% gain in model recall**, setting new performance benchmarks
- Built a **Business Intelligence Summariser** leveraging structured sales and finance data to project it to a **unified analytics space**, generating concise revenue summaries and providing insightful responses to **10k+ business queries**.

Machine Learning Engineer @ LTI Mindtree, India

Dec 2020 – Dec 2022

- Led a **3-member engineering team** to architect and deploy a **geo-distributed climate analytics system** from scratch, integrating multimodal deep learning across **edge, HPC, and cloud**, delivered **end-to-end within 6 months**
- Reduced **model training latency by 35%** through **GPU acceleration (CUDA)**, distributed **TensorFlow** pipelines, and **AWS SageMaker**, enhancing prediction throughput by **20%** and compute efficiency by **15%**.
- Incorporated **Federated Learning** with lightweight **ML algorithms (XGBoost)** on low-resource edge nodes, achieving **97% rainfall prediction accuracy** while ensuring data privacy and scalable **MLOps** workflows.
- Engineered **multi-variable time-series forecasting pipelines** using **LSTM** and **PyTorch**, processing **100+ years** of oceanographic and weather data across **10+ climate variables**, improving long-horizon prediction stability by **22%** and setting new evaluation benchmarks across regional climate datasets.

Skills

Generative AI & LLMs:	OpenAI GPT-4/4o, Azure OpenAI, Claude, LangChain, LangGraph, LlamaIndex, AutoGen, CrewAI, RAG, Prompt Engineering, Agentic AI, AI Evaluations, Hallucination Reduction, Gradio, Hugging Face Transformers
Machine Learning & Deep Learning:	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, OpenCV, NLP, Deep Learning, Time-Series Forecasting, LSTM, SHAP, LIME, Federated Learning, CUDA, CLIP, ARIMA, Prophet
Vector Databases & Semantic Search:	FAISS, Pinecone, ChromaDB, Sentence Transformers, Semantic Search, Embeddings, CrossEncoder Re-ranking, BM25, CLIP Embeddings
Cloud & MLOps:	Azure ML, Databricks, AWS SageMaker, Docker, Kubernetes, MLflow, Azure Data Factory, Delta Lake, Kafka, CI/CD Pipelines, Experiment Tracking, Model Monitoring, Azure Event Hubs, PySpark
Backend Engineering & Infrastructure:	FastAPI, Flask, REST APIs, gRPC, Git, Linux, Async APIs, Microservices, Event-Driven Systems, Snowflake
Languages:	Python, C/C++, Java, R, SQL, Shell, JavaScript, Pandas, NumPy, Jupyter

Projects

Intelligent Financial Document Analyzer | LangChain, FAISS, FastAPI, Docker, RAG Evaluation

- Architected an end-to-end **RAG pipeline** for contextual **financial document understanding** using **semantic retrieval** and **transformer-based reasoning**.
- Integrated **entity extraction** and **prompt optimization** techniques to improve **retrieval quality** and downstream **reasoning accuracy**.
- Added **evaluation pipelines** for **retrieval precision**, **semantic relevance**, and **hallucination reduction** using **structured test datasets** and **automated scoring workflows**.
- Launched scalable **inference APIs** using **FastAPI** and **Docker** for enterprise **document analysis workflows**.

Real-Time Fraud Detection System | Kafka, PySpark, XGBoost, FastAPI

- Developed a **streaming fraud detection pipeline** for continuous **transaction monitoring** using **Kafka** and **machine learning-based anomaly detection**.
- Crafted **low-latency inference APIs** with **behavioral** and **temporal feature engineering** for **scalable fraud prediction**.

Multi-Model AI Retrieval System | CLIP, FAISS, Sentence Transformers

- Assembled unified **text-image embedding pipelines** for **semantic retrieval** using **CLIP** and transformer-based embedding models.
- Implemented **hybrid retrieval** with **FAISS** and **BM25 re-ranking** for improved **cross-modal search** relevance.

MLOps Pipeline for Credit Risk Prediction | MLflow, XGBoost, Docker

- Developed an **end-to-end ML pipeline** covering **preprocessing**, **training**, **model versioning**, and **deployment workflows**.
- Established reproducible **experimentation** and **inference APIs** using **MLflow**, **FastAPI**, and **Docker**.

Time-Series Forecasting | ARIMA, Prophet, LSTM

- Constructed **forecasting models** for **climate** and **financial datasets** using **statistical** and **deep learning** approaches.
- Performed **feature engineering**, **stationarity analysis**, and **walk-forward validation** for **robust long-horizon forecasting**.

OPEN SOURCE & PORTFOLIO

- Maintained hands-on GitHub repositories for RAG pipelines, streaming ML systems, and production-ready FastAPI deployments.
- Built reproducible AI workflows with clear documentation, modular architecture, deployment instructions, and evaluation reports for experimentation.
- Created architecture diagrams and deployment documentation for distributed AI systems including RAG pipelines, vector retrieval systems, and streaming inference APIs.

RESEARCH & LEADERSHIP

- Conducted thesis research on SEMLHI, a software engineering methodology for machine learning systems in health informatics.
- Mentored junior engineers and interns on model experimentation, feature engineering, and ML workflow development.
- Collaborated with cross-functional analytics and engineering teams on enterprise AI adoption and scalable ML deployment.
- Built multiple independent projects focused on Generative AI, RAG pipelines, and production-grade AI systems.